Contact

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www.linkedin.com/in/tarunsha (LinkedIn)
shtarun.github.io (Personal)

Top Skills

Machine Learning
Audio Processing
Linux Kernel

Languages

English (Full Professional)
Hindi (Native or Bilingual)
German (Elementary)

Certifications

Machine Learning
Object-Oriented Design
Convolutional Neural Networks
Introduction to Git and GitHub
Software Architecture

Honors-Awards

JEE Advanced
JEE Main
Asha Khanna Award
JLR Global Digital Delivery Hackathon ‘21

Publications

Primary suspension failure analysis in FIAT type LHB bogies and life estimation

Tarun Sharma

Building Twinity Labs | ex JLR | IIT Kanpur | Dual Degree
Bengaluru, Karnataka, India

Summary

Building Digital Twin ecosystems for the future ;)

Worked with JLR to build the futuristic Software Defined Vehicles and Global Testing Platform with many other projects and initiatives.

I perceive and enjoy the beauty at intersection of different fields not just restricted to science. I have been involved in a diverse range of projects which has helped me gain deep knowledge on various domains relating to computer science, mechanical engineering and psychotherapy.

A Dual Degree graduate student from the Mechanical Engineering department of IIT Kanpur.

Experience

Twinity Labs
Founder & CEO
September 2025 - Present (9 months)

Jaguar Land Rover India
5 years

Senior Software Engineer, Software-Defined Vehicles
January 2025 - August 2025 (8 months)
Bengaluru, Karnataka, India

• Open-source-first IaC & build stack: Triggered an enterprise-wide shift from commercial Terraform to OpenTofu, refactoring shared modules, remote state backends, and CI workflows; integrated open-source platform enablers such as Lorry (artifact promotion pipelines), Trustable Software Framework (policy-as-code, SBOM & compliance checks), and BuildBarn-backed remote builds/caching to cut licensing spend, harden supply-chain security, and make infra and build environments reproducible from a single declarative stack.

• Zero-downtime platform migration: Orchestrated Software Integration Factory's decommissioning, migrating 450+ GitLab repos, Podman base images, and infra elements into JLR's ecosystem - kept Central Compute Module builds green and teams unblocked during the cutover.
• Scalable cloud footing: Coordinated with Digital to provision a fresh AWS account, aligning security, access, and cost controls with program growth.
• Platform-agnostic pipelines: Released MR1 pipeline with SoC/MCU parity - shrinking duplication and accelerating reuse across programs.
• Frictionless application onboarding: Published a standardized repository template so new apps can integrate with the SDV platform (Central Compute Module's SoC/MCU) in hours instead of days.
• Hardware readiness on new silicon: Rebuilt and validated the CCMA MCU integrated image for the NXP S32N board under CI, ensuring performance and compatibility before scale-up.
• Governance & velocity: Secured alignment from senior stakeholders on SDV GitLab restructuring, clarifying ownership and streamlining end-to-end delivery.
• Hands-free device operations: Enabled automated flashing in MR1 & MR2 for Central Compute Module's MCU, removing manual steps and tightening feedback loops.

Innovation Champion - Open Innovation Hub
December 2024 - August 2025 (9 months)

• Championed cross-functional innovation by identifying critical business gaps within JLR and driving end-to-end startup collaborations via Plug and Play's global innovation platform. Responsible for owning the full lifecycle - from problem discovery and startup scouting to proof-of-concept (PoC) execution and funding alignment.
• Independently identified a strategic opportunity in Digital Twinning and led a high-impact engagement with IndusTANTRA (founded by Prof. N.S. Vyas, IIT Kanpur) - India's foremost authority in the domain - bypassing traditional scouting channels and reducing engagement turnaround from months to under 4 weeks by directly aligning with senior JLR leadership and initiating ML-driven surrogate modelling projects to enhance JLR's Digital Twin ecosystem

Software Engineer, Software-Defined Vehicles
November 2023 - February 2025 (1 year 4 months)
Bengaluru, Karnataka, India

• Executive-facing delivery: Presented multiple times to the Boards of JLR and Tata Motors on base OS/platform progress, aligning leadership on scope, risks, and milestones.

• Build ecosystem you can trust: Maintained the CCMA SoC MR2 repo; integrated Podman via BuildStream and standardized library builds with BuildStream, CMake, Ninja, Meson - raising build reliability and developer throughput.
• OS/BSP bring-up lead: Led board bring-up for OS and BSP integration, configuring Linux kernel, network interfaces, and systemd services to achieve stable, repeatable boots on target hardware.
• Deterministic platform services: With Codethink, implemented a Deterministic Construction Service for JLR’s custom Linux OS - improving determinism and supporting safety goals.
• Connected, production-ready stack: Integrated AWS, Alexa, and RTI DDS, extending connectivity, telemetry, and inter-process communication for in-vehicle and cloud flows.
• Boot performance observability: Added bootchart generation + KPIs in CI; built an automated AWS (S3, Lambda, EC2, Redshift) workflow that flashes physical SoCs, generates bootcharts, and feeds Tableau dashboards for data-driven regressions.
• Safety groundwork toward ISO 26262: Contributed to the fault manager architecture and attempted to build an independent Safe Linux prototype (custom SCHED_DEADLINE kernel + mock deadlineed processes) to aid functional-safety assessment.
• ASIL B leadership: Led a cross-functional effort targeting ASIL B certification for a safety-certified Linux distribution, aligning practices to ISO 26262.
• Image + SDK shipped with partners: Co-delivered integrated image and SDK with Codethink, smoothing downstream onboarding and iteration.
• People development: Mentored 4 interns (2 directly) on Rust vs C++ trade-offs (outcomes led to PPOs), coached and directly managed 2 fresh software engineering graduates, lifting team velocity and confidence.

Software Engineer, Global Testing Platform

September 2022 - February 2024 (1 year 6 months)

• Worked on Automated Keyword-Driven Global Testing Platform to create a robust and unified testing platform for JLR
• Architected and developed a Python based Test Executable Generator, by factory and builder designs, enabling end-to-end automated test execution across various toolchains (dSpace, Vector, National Instruments) on 300+ global test assets
• Developed "Mars Attacks," a FastAPI microservice for processing GCP data, enhancing visibility of test environment setup

• Streamlined CI/CD pipelines, implementing validation tests and automated test environment cleanup to reduce infrastructure costs

Graduate Software Engineer Trainee

September 2020 - August 2022 (2 years)

Bengaluru, Karnataka, India

• Engaged in predictive maintenance research using LSTM, focusing on spectrograms, clustering, and image processing. This concept is now with the patent office.
• Collaborated on Virtual & Extended Reality experiences, presenting innovative solutions to the Board of Directors
• Contributed to developing data pipelines, machine learning models for audio signal analysis, and initiating Software-in-Loop testing processes
• Provided technical mentorship, participated in hiring processes for full-time roles and internships, and acted as a liaison with IIT Kanpur for recruitment initiatives

Indian Institute of Technology, Kanpur

9 months

Project Engineer

July 2020 - September 2020 (3 months)

Worked for a brief period on a project by Hyper Poland(currently Nevemo), involving the prototype of a MagRail, by framing the problem statement and approach for the solution to the project.

Teaching Assistant

January 2020 - June 2020 (6 months)

For the course Engineering Graphics(TA101A) under Prof. Anupam Saxena and Prof. Bharat Lohani, consisting of 40 undergraduate students.

Association of Mechanical Engineers, IIT Kanpur

President

August 2019 - June 2020 (11 months)

• Arranged training sessions for the department from MathWorks and Ansys on dynamic and static solid simulations and FSI simulations
• Conducted multiple casual technical engagements for the students of the department
• Managed a small budget of INR 3 lacs for the society by working with the department administration

Indian Institute of Technology, Kanpur
4 months

Motion Planning of a 6DOF Robot Arm Manipulator
August 2019 - November 2019 (4 months)

Project Supervisor: Dr. Ashish Dutta

Course: Robot Motion Planning

• Used MATLAB to construct the C-Space of a 3DOF & 2DOF Robot arm with obstacles.

• Used sampling-based planners (A* Algorithm) for the path planning of a mobile robot in presence of obstacles.

• Used Rapidly Exploring Random trees for path planning of a mobile robot.

Teaching Assistant
August 2019 - November 2019 (4 months)

For the course Advanced Mechanics of Solids (ME321A) under Prof. Ishan Sharma, consisting of 100+ undergraduate students.

FEA of Railway Couplers used in LHB Coaches
August 2019 - September 2019 (2 months)

• Provided Frontier Technologies Pvt. Ltd. with the Finite Element Models of Railway Couplers used in LHB Bogies in India.

• Used ANSYS to verify the CAD model of the coupler assembly and performed FEA for various loading conditions like Buff, Draft & Bending to check the equivalent Von-Mises stress distribution, total deformation and directional deformation of the coupler.

• Simulated the Anti Climbing Feature of the coupler assembly in ANSYS under 45 tonnes of force.

Counselling Service, IIT Kanpur
Orientation Team Member
July 2019 - August 2019 (2 months)

IIT Kanpur

Assisted core team in the smooth conduction of 8-day long orientation program for 1000+ postgraduate students

Indian Institute of Technology, Kanpur
10 months

Algorithm Study of Constrained and Unconstrained Optimization Techniques
March 2019 - April 2019 (2 months)

Project Supervisor: Prof. Bhaskar Dasgupta

Course: Optimization Methods

• Analysed the performance of unconstrained optimization methods like Steepest Descent, DFP, BFGS and Powell's Conjugate Direction.
• Studied the effect of different parameters like initial point, modality and concluded the global convergence on 50 different test functions.
• Linear and Quadratic Programming with Simplex, Active Set method and Lemke's set method for multiple constraints was studied.

Design & Development of an Electromagnetic Actuator

August 2018 - April 2019 (9 months)

B.Tech. Project

Project Supervisor: Prof. Mohit Law

• Developed a cheap Moving Iron Type Linear Electromagnetic Actuator to be used as active damping devices
• Optimized the spatial and functional parameters using COMSOL & FLUX to achieve higher force to volume ratio
• Designed a low friction linear Flexural Bearing on DS Solidworks and carried out its fatigue analysis
• Isolated the natural frequency and obtained a flat frequency response in 30-140 Hz

Structure Integrated Sensors and Actuators to Monitor and Renew Machine Tool Deformance

December 2018 - February 2019 (3 months)

Position: Student Research Associate

Project Supervisor: Prof. Mohit Law

• Designed a Tuned Mass Damped Boring Bar and a Compression Holder using DS Solidworks
• Used CutPro for modal testing of the developed Tuned Mass Damped Boring Bar and an Eddy Current Damped Boring Bar
• Optimized the mass and stiffness of the tuned mass using Maple & MATLAB for chatter elimination and higher Material Removal Rate

Modelling of Railway Vehicle Dynamics: A Multi-Body Analytical Approach

July 2018 - December 2018 (6 months)

Position: Student Research Associate

Project Supervisor: Prof. N.S. Vyas

• Determined contact patch co-ordinates as a function of lateral perturbation by solving the kinematic equations
• Iteratively solved the equations of motion on MATLAB to get forces and estimate critical speed of stability
• Observed response characteristics of a railway coupler model by varying source frequency, spring stiffness, draft gear friction and coupler slack
• Modelled a rail-wheel pair in Simpack and observed the motion on a straight & a curved track by varying the wheel positions
• Imparted uncertainties like Camber, Yaw and Toe-in to the wheelset and used Simpack Post to visualise its motion behaviour

Output feedback stabilization of inverted pendulum

September 2018 - November 2018 (3 months)

Project Supervisor: Prof. Ramprasad Potluri

Course: Basics of Modern Control

• Implemented the research paper, 'Output feedback stabilization of inverted pendulum on a cart in the presence of uncertainties'
• Investigated the singularly perturbed form of the closed loop system including Extended High Gain Observers.
• Used MATLAB and Visio to simulate and solve the steady state feedback equations and for Block Diagram visualization.

Helicopter Coupled Trim Analysis using MATLAB for a UH-60A Black Hawk helicopter

August 2018 - November 2018 (4 months)

Project Supervisor: Dr. Abhishek

Course: Helicopter Dynamics & Aeroelasticity

• Programmed a numerical solution to the combined blade-element/momentum theory (BEMT)
• Used Newmark's algorithm to solve the flap response equation numerically in a coupled trim solution.
• Calculated all the blade hub shear forces and moments in the rotating frame of reference as a function of azimuth for a uniform inflow.
• Performed the free flight coupled trim analysis for the same helicopter and found the variation of control angles, vehicle shaft angles and non-dimensional mean hub loads vs forward speed

ETA Technology Pvt. Ltd.
Summer Internship
May 2018 - July 2018 (3 months)
Bengaluru

• Attempted to design and develop an axially frictionless Hydrostatic Bearing for withstanding 60000 kgf of radial force
• Designed components of a Friction Welding Machine using Solidworks and AutoCAD
• Optimized parameters of an Electrical Upsetting and Metal Gathering Machine to get the desired valve profile
• Fixed, redesigned and carried out failure analysis of a broken Chamfering tool

R&D, Tata Steel
Summer Internship
May 2017 - July 2017 (3 months)
Jamshedpur

• Investigated uni-axial tensile testing theory focusing on the plastic region of deformation for high strain rates
• Modelled a tensile specimen using Abaqus & learnt about the uni-axial and bi-axial tensile testing machines
• The tensile test results contributed in designing new specimen for specific tests for automotive applications

Indian Institute of Technology, Kanpur
Design & Development of a Document Shredder & an Automated Box Shifting Mechanism
August 2016 - April 2017 (9 months)

Project Supervisors: Prof. S.K. Choudhary, Prof. Rajiv Shekhar

Courses: TA201A & TA202A

• Deployed SolidWorks 2016 and AutoCAD 2015 for modelling the mechanisms.
• Designed and fabricated the working prototypes using CNC machine, Lathe machine and Drilling machine.
• Made a Costing Report, Business Plan and Marketing Brochure for the developed prototypes of the mechanisms.

Entrepreneurship Cell, IIT Kanpur
Senior Executive
September 2016 - March 2017 (7 months)

Indian Institute Of Technology - Kanpur

Senior Executive at Startup Internship Program,E-Cell, IIT Kanpur

• Piloted a 5 membered team to organize Summer'17 Internship program while managing the website and logistics
• Contacted 200+ companies to take part in Start-up Internship Program 2017 for various profiles in core and non- core departments

Education

Indian Institute of Technology, Kanpur

BT MT Dual Degree · (2015 - 2020)

Loyola School Jamshedpur

Mathematics and Computer Science · (2002 - 2015)